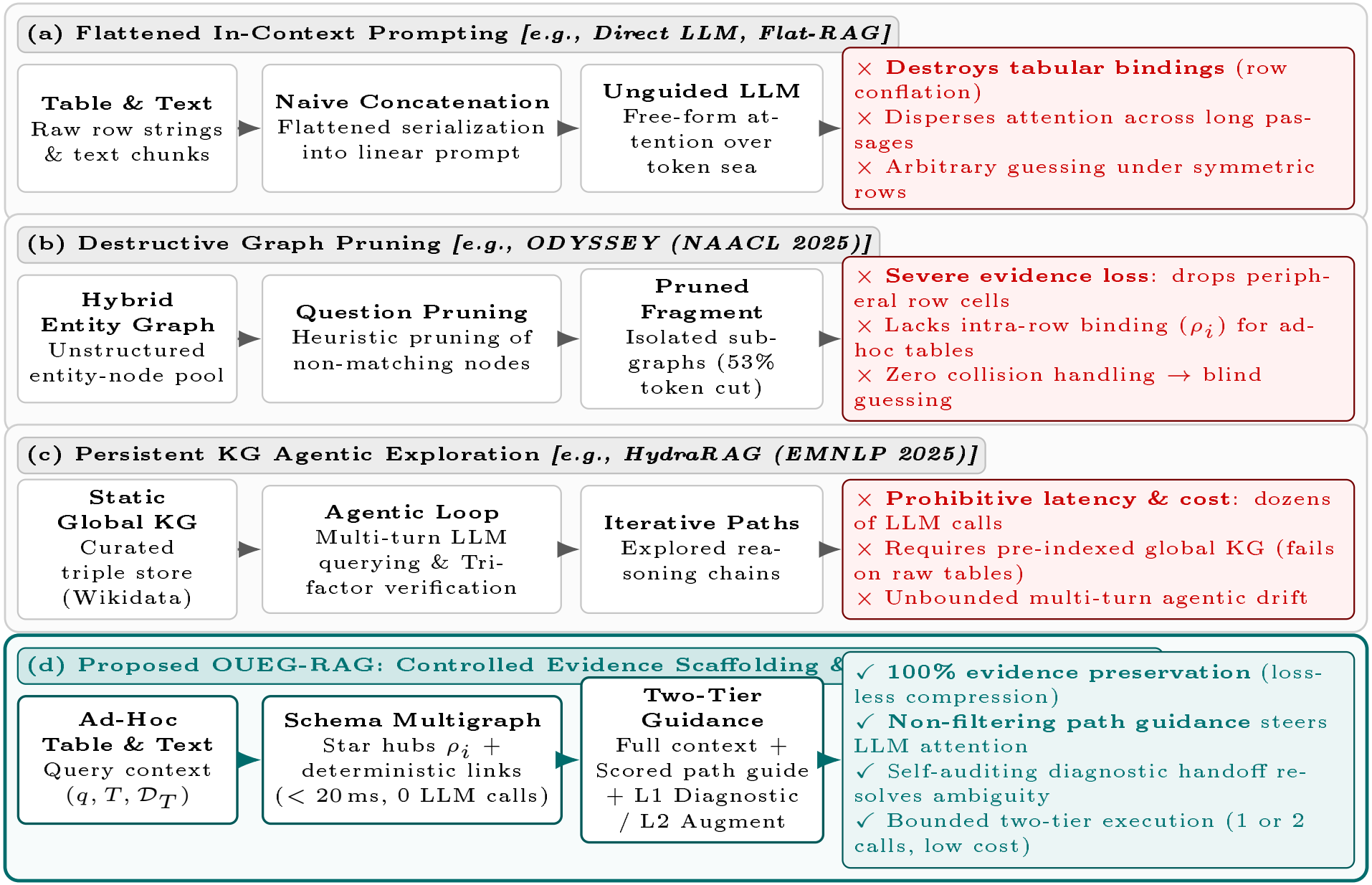

(a) Flattened In-Context Prompting [e.g., Direct LLM, Flat-RAG]
Table & Text
Raw row strings & text chunks
Naive Concatenation
Flattened serialization into linear prompt
Unguided LLM
Free-form attention over token sea
× Destroys tabular bindings (row conflation)
× Disperses attention across long passages
× Arbitrary guessing under symmetric rows
(b) Destructive Graph Pruning [e.g., ODYSSEY (NAACL 2025)]
Hybrid Entity Graph
Unstructured entity-node pool
Question Pruning
Heuristic pruning of non-matching nodes
Pruned Fragment
Isolated subgraphs (53% token cut)
× Severe evidence loss: drops peripheral row cells
× Lacks intra-row binding (ρ_i) for ad-hoc tables
× Zero collision handling → blind guessing
(c) Persistent KG Agentic Exploration [e.g., HydraRAG (EMNLP 2025)]
Static Global KG
Curated triple store (Wikidata)
Agentic Loop
Multi-turn LLM querying & Trifactor verification
Iterative Paths
Explored reasoning chains
× Prohibitive latency & cost: dozens of LLM calls
× Requires pre-indexed global KG (fails on raw tables)
× Unbounded multi-turn agentic drift
(d) Proposed OUEG-RAG: Controlled Evidence Scaffolding &
Ad-Hoc Table & Text
Query context $(q, T, \mathcal{D}_T)$
Schema Multigraph
Star hubs ρ_i + deterministic links (< 20 ms, 0 LLM calls)
Two-Tier Guidance
Full context + Scored path guide + L1 Diagnostic / L2 Augment
✓ 100% evidence preservation (lossless compression)
✓ Non-filtering path guidance steers LLM attention
✓ Self-auditing diagnostic handoff resolves ambiguity
✓ Bounded two-tier execution (1 or 2 calls, low cost)